How-To Library

Informatica

Configuring Enterprise Information Catalog on Amazon Web Services

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Abstract

You can verify the default configuration details and Elastic IP address requirements before installing and deploying Enterprise Information Catalog on Amazon Web Services (AWS), After installing and deploying, you can monitor the progress of creating the cluster instance and Informatica domain. To perform advance configuration tasks, you must complete the prerequisites associated with the tasks. The log files help you troubleshoot issues that you might encounter

It is assumed that you are familiar with Amazon Web Services and the service management tasks performed on the AWS Management Console.

Supported Versions

- Enterprise Information Catalog 10.1.1

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Overview

Enterprise Information Catalog brings together all data assets in an enterprise and presents a comprehensive view of the data assets and data asset relationships. A data asset is a type of data object, such as a physical data source, HDFS, or big-data repository. The data assets in the enterprise might exist in relational databases, purpose-built applications, reporting tools, HDFS, and other big-data repositories.

Enterprise Information Catalog captures the physical and operational metadata for a large number of data assets that you use to determine the effectiveness of enterprise data. Metadata is data about data. Metadata contains details about the structure of data sources. Metadata also includes information, such as data patterns, data types, relationships between columns, and relationships between multiple data sources.

Enterprise Information Catalog gathers information related to metadata across the enterprise. The metadata includes column data statistics, data domains, data object relationships, and data lineage information. A comprehensive view of enterprise metadata can help you make critical decisions on data integration, data quality, and data governance in the enterprise.

You can install and deploy Enterprise Information Catalog on AWS. Enterprise Information Catalog is available as Software as a Service (SaaS) in Amazon Marketplace. The installation and deployment of Enterprise Information Catalog on AWS provides the following benefits compared to an on-premise installation and deployment:

- Easy and quick deployment.
- Minimal manual configuration.

- Sample demo resources to jump start the first use.
- Predefined single node or multiple-node embedded cluster sizes based on deployment requirements.
- Sandbox environment for testing.
- Licensing cost based on deployment requirements.

Installing and deploying Enterprise Information Catalog on AWS provides you the following features:

- Catalog Service on an embedded cluster.
- Scanner resources for all metadata sources.
- Hive engine for profiling.
- Single-node Informatica domain on Red Hat Enterprise Linux version 7.2.

Default Configuration Details

The following are the default configuration details when you install and deploy Enterprise Information Catalog on AWS:

Base Amazon Machine Image Specifications

All Amazon Machine Images (AMI) include the following specifications:

Specification	Type
Operating System	Red Hat Enterprise Linux 7.2 (HVM)
Volume Type	SSD
Cluster Node Image	150 GB EBS volume
Volume Type	Provisioned IOPS
IOPS	4000

Domain Instance Specifications

Specification	Type
Domain Node Image	225 GB EBS volume
Volume Type	Provisioned IOPS
IOPS	4000

Database Configuration Details

The database configured is RDS for SQL Server Standard Edition with 200 GB storage.

The following table lists the values configured for the database parameters:

Parameter	Value
DB instance class	db.m4.2xlarge
License ModelLicense model information for this DB instance	License Included
Storage Type	General Purpose (SSD)
IOPS Fixed number of I/O Operations Per Second (IOPS) supported by this DB Instance	disabled

Amazon EC2 Configuration

Specification	Type
Cluster Node Image	150 GB
Cluster Node Image Type	c4.8xlarge
Domain Node Image	225 GB
Domain Node Image Type	- c4.8xlarge - c4.4xlarge
Volume Type for both cluster node and domain node	Provisioned IOPS

Elastic IP Address Requirements

As part of the free tier, Enterprise Information Catalog on AWS supports five Elastic IP addresses for an AWS account. Enterprise Information Catalog uses one Elastic IP address for each node in the cluster and one Elastic IP address for the Informatica Administrator Server.
The following are the requirements for Elastic IP addresses for clusters of different sizes:

Cluster Size	Number of Nodes	Elastic IP Addresses Required
Small	1	2
Medium	3	4
Large	6	7

Monitoring Instance Provision and Informatica Domain Creation

You can monitor the progress of creating the cluster instance and Informatica domain.

1. Select the stack that you are creating, then select the **Events** tab.
2. When the stack creation is complete, select the **Resources** tab.

   The **Resources** tab displays information about the stack and Big Data Management instance. You can select the linked Physical ID properties of resources to get more information about them.
3. Click the **Outputs** tab.

When the Informatica domain setup is complete, the **Outputs** tab displays the following information:

Property	Description
ICSMultiNodeClusterURL	URL to access the Informatica Cluster Service Hadoop gateway node. If you had selected the cluster size as *small*, the **Output** tab displays the *ICSSingleNodeClusterURL* property name.
InstanceID	Hostname of the Informatica domain.
InstanceSetupLogs	Location of the setup log files for the Informatica domain EC2 instance.
InformaticaAdminConsoleURL	URL of Informatica Administrator. Use the Administrator tool to administer Informatica services.
InformaticaAdminConsoleServerLogs	Location of the Informatica Administrator log files.

Note: If the **Outputs** tab is not populated with this information, wait for domain setup to be complete.

4. Open the **Resources** tab. Click the **Physical ID** of the `AdministrationServer` property. The **Physical ID** corresponds to the name of the Informatica domain.

   The **Instance Administration** screen opens.

   You can use the **Instance Administration** screen to launch the Enterprise Information Catalog instance. You can also get information such as the public DNS and public IP address.

Prerequisites for Advance Configuration Tasks

The advance configuration tasks include administration tasks that you can perform after installing and deploying Enterprise Information Catalog on AWS. You must complete the prerequisites using the AWS Management Console before you perform the advance configuration tasks.

The following are the advance configuration tasks for which you must complete the prerequisites:

- Creating new resources.
- Adding nodes to the Informatica Cluster Service.
- Deleting nodes.

Prerequisite for Adding New Resources

Before adding new resources, make sure that the resource and AWS are included in the same security group.

Prerequisites for Adding Nodes

Perform the following steps in the AWS Management Console before you add nodes to the Informatica Cluster Service:

1. Launch the instance using the predefined AMI. Make sure that you use the AMI corresponding to the region from where you are launching the instance. Make sure that you follow the points listed below before selecting the instance:
   a. Select the same instance type that you had selected while creating the cluster.
   b. Edit the security group and add the new instance to the existing cluster security group.
   c. Select the existing keypair that you might have given while creating the stack using Enterprise Information Catalog CloudFormation template.

2. Create an Elastic IP address from AWS Management Console.
3. Associate the Elastic IP address with the instance you launched.

   The AWS Management Console provides you a new public DNS name.
4. Log in to the newly created instance using Putty with the keypair you selected.
5. Set the public DNS name as hostname for the newly created instance by running the following commands as a *root* user in the instance:
   a. sed --in-place '1s/.*/$publicDnsName/' /etc/hostname. Alternatively, you can replace the first line of the `/etc/hostname` file with `publicDnsName`
   b. echo HOSTNAME=$publicDnsName >> /etc/sysconfig/network. Alternatively, you can add `HOSTNAME=$publicDnsName` in `etc/sysconfig/network`
   c. sed -i '$ a preserve_hostname: true' /etc/cloud/cloud.cfg. Alternatively, you can append the line `preserve_hostname: true` to the `/etc/cloud/cloud.cfg` file.
6. Run the following commands as *root* user to configure Java to run scanners:
   a. ln -s /usr/jdk64/jdk1.8.0_60/bin/java /etc/alternatives/java &> /dev/null
   b. ln -s /etc/alternatives/java /usr/bin/java &> /dev/null
7. Add the private IP address and public DNS name of the instance to `/etc/hosts` of all machines in the cluster and the machine that runs Informatica Administrator. You need not create an alias name in `/etc/hosts` file.
8. Add the private IP address and the public DNS name of other machines to the instance. You can copy the content of `/etc/hosts` from one of the machines in the cluster to the instance.
9. Reboot the instance for the changes to take effect.
10. Add the public DNS name of the instance to the Informatica Cluster Service using Informatica Administrator as a Hadoop node. See the *Informatica Administrator Reference for Live Data Map* guide for more information.

All prerequisite packages are installed and passwordless SSH is enabled for the instance.

AMI Reference

The following are the AMIs for all regions along with the configuration details:

Region	INFAEICADMINHVM Value	INFAEICCLUSTERHVM Value	Multiple Availability Zones Region Mapping Supported (Yes/No)
ap-northeast-1	ami-42b0f725	ami-cbbef9ac	Yes
ap-northeast-2	ami-d42ffeba	ami-3d2ffe53	Yes
ap-south-1	ami-07790868	ami-5f7c0d30	Yes
ap-southeast-1	ami-d0f246b3	ami-c2f246a1	No
ap-southeast-2	ami-6f73750c	ami-02717761	Yes
ca-central-1	ami-37c17c53	ami-acc37ec8	Yes
eu-central-1	ami-ebd91084	ami-ead91085	No
eu-west-1	ami-d4f6aeb2	ami-e2e8b084	Yes

Region	INFAEICADMINHVM Value	INFAEICCLUSTERHVM Value	Multiple Availability Zones Region Mapping Supported (Yes/No)
eu-west-2	ami-14445170	ami-2b584d4f	Yes
sa-east-1	ami-86781cea	ami-72791d1e	Yes
us-east-1	ami-cff900d9	ami-c0f900d6	Yes
us-east-2	ami-0b3a1f6e	ami-7a3a1f1f	Yes
us-west-1	ami-b59dc0d5	ami-f79ec397	No
us-west-2	ami-89c47fe9	ami-71c57e11	Yes

- INFAEICADMINHVM represents the administrator server image for Enterprise Information Catalog on AWS.
- INFAEICCLUSTERHVM represents the Informatica Hadoop Cluster image for Informatica Cluster Service.

Prerequisites for Deleting Nodes

Complete the following prerequisites before deleting a node from the Informatica Cluster Service:

1. Delete the node from Informatica Administrator and recycle Informatica Cluster Service.
2. After Informatica Cluster Service is up and running, remove the Elastic IP address associated with the deleted node from AWS Management Console.
3. Delete the EC2 instance for the deleted node from the AWS Management Console.

Troubleshooting

You can see the following log files for more information about any errors that might occur:

Log File Location	Description
/var/log/cfn-init.log	Log files related to the EC2 instances.
/installation.log	Log files related to the installation.
/opt/informatica/logs/testNode/	Log files related to Informatica services.

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